Cluster-State Unitary Decomposition with Native Free Evolution and SNAP

Codex
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We study how much of the cluster-state transfer unitary in a dispersive transmon-cavity system can be implemented by native free evolution alone, and how that answer changes once photon-number selective phase gates (SNAP) are added. The logical target acts on $\{|g,0\rangle, |g,1\rangle, |e,0\rangle, |e,1\rangle\}$ and is $U_{\rm cl} = \mathrm{SWAP} \cdot \mathrm{CZ} \cdot (H \otimes I)$. The baseline gate set is restricted to displacement, qubit rotations, and a wait-only entangling primitive with optimized duration. Extended families add SNAP either as a final correction layer, interleaved after each free-evolution block, or in both roles. Using `cqed_sim` for all synthesis and diagnostics, we find that native free evolution is indispensable: removing it from the best native sequence drops the logical fidelity to 0.167, and removing it from the best SNAP-assisted sequence drops the fidelity to 0.322. SNAP is therefore not a replacement entangler. It is also not purely redundant. The best no-SNAP family reaches fidelity 0.964 with six free-evolution blocks, depth 20, and total wait time 1018 ns. The best SNAP-extended family reaches 0.988 with six free-evolution blocks and six interleaved SNAP layers at essentially the same wait budget (1027 ns), indicating that SNAP mainly supplies explicit cavity-only phase correction rather than reducing the native entangling load at the highest-fidelity point. At a practical threshold of 0.95 fidelity, however, SNAP reduces the required entangling resources: the interleaved family reaches 0.964 with four free-evolution blocks, depth 18, and 769 ns total wait time, whereas the no-SNAP family first exceeds 0.95 only at six blocks, depth 20, and 1018 ns. A follow-up improvement pass then re-optimizes the two depth-6 winners at $N_{\rm cav} = 6$, raising the native case from replay fidelity 0.9177 to 0.9234 and the interleaved-SNAP case from 0.9473 to 0.9908. Coherent model-backed replay at $N_{\rm cav} = 8$ gives 0.9238 and 0.9867, while a documented Lindblad noise surrogate yields mean probe-state fidelities 0.8907 and 0.9351, respectively. Tail-only SNAP layers provide little benefit.

I. INTRODUCTION

Cluster-state generation by repeated application of a fixed transfer unitary is a standard route to measurement-based quantum computation [1]. In circuit QED, the natural hardware platform is a dispersively coupled qubit-oscillator pair, where native dispersive waiting, cavity displacements, qubit rotations, and oscillator phase gates all provide different pieces of the required control toolbox [2–5]. The key design question here is narrower than full universality: when the target is the four-dimensional cluster-state transfer unitary, which part of the job is done by native free evolution and which part genuinely benefits from explicit number-dependent phase programming?

Earlier work in this repository established that the cluster target can be implemented either by broader selective gate libraries or by direct optimal control, but that result did not isolate the role of the wait-only entangler. The present study does. We first restrict the search to $\{D, R_q, U_{\rm FE}(t)\}$, where $U_{\rm FE}(t)$ is a no-drive free-evolution conditional-phase gate with optimized wait time. We then compare against extended families that add SNAP layers in different positions. This lets us answer three concrete questions: does SNAP reduce gate depth, does it reduce the total entangling wait time needed for useful fidelities, and does it add independent functionality or mostly redundant phase freedom?

II. SYSTEM AND METHODS

A. Hamiltonian

We use the dispersive transmon-cavity model already implemented in `cqed_sim` [6]. In the rotating frame used by the synthesis stack,

$$\frac{H}{\hbar} = \omega_c a^\dagger a + \omega_q b^\dagger b + \frac{\alpha}{2} b^{\dagger 2} b^2 + \chi\, a^\dagger a\, b^\dagger b + \chi' a^{\dagger 2} a^2 b^\dagger b + \frac{K}{2} a^{\dagger 2} a^2, \tag{1}$$

with a the cavity annihilation operator, b the transmon lowering operator, χ the dispersive shift, χ' the first higher-order dispersive correction, and K the cavity self-Kerr. The logical basis is

$$\mathcal{B}_{\rm log} = \{|g,0\rangle, |g,1\rangle, |e,0\rangle, |e,1\rangle\}, \tag{2}$$

and the target unitary is

$$U_{\rm cl} = \mathrm{SWAP} \cdot \mathrm{CZ} \cdot (H \otimes I), \tag{3}$$

equivalently `make_target("cluster", n_match=1)`.

B. Simulation Parameters

TABLE I. System and nominal gate parameters used in the decomposition study.

Parameter	Value	Unit
$\omega_q/2\pi$	6.150	GHz
$\omega_c/2\pi$	5.241	GHz
$\alpha/2\pi$	-255	MHz
$\chi/2\pi$	-2.84	MHz
$\chi'/2\pi$	-21	kHz
$K/2\pi$	-28	kHz
$N_{\rm tr}$	2	levels
$N_{\rm cav}$ (search)	4	Fock states
$D(\alpha)$ duration	80	ns
$R_q(\theta, \phi)$ duration	40	ns
SNAP duration	200	ns
FE wait bounds	40–400	ns

C. Computational Approach

All searches use the `cqed_sim.unitary_synthesis` stack [6]: `Displacement`, `QubitRotation`, `FreeEvolveCondPhase`, `SNAP`, `GateSequence`, `TargetUnitary`, `Subspace`, and `UnitarySynthesizer`. The optimization objective combines unitary fidelity and leakage penalty, with a small duration-weight sweep used only for compactness diagnostics. We use a two-stage workflow: a coarse screening pass with 20 iterations and seed 17, followed by refinement of the best case in each family with 35 iterations and seeds 17 and 42.

Four families were searched:

$$\mathcal{G}_{\rm FE} : [D\,R\,{\rm FE}]^b DR, \tag{4}$$

$$\mathcal{G}_{\rm tail} : [D\,R\,{\rm FE}]^b D\,{\rm SNAP}\,R, \tag{5}$$

$$\mathcal{G}_{\rm int} : [D\,R\,{\rm FE}\,{\rm SNAP}]^b DR, \tag{6}$$

$$\mathcal{G}_{\rm hyb} : [D\,R\,{\rm FE}\,{\rm SNAP}]^b D\,{\rm SNAP}\,R. \tag{7}$$

Here b is the number of free-evolution blocks. The baseline family $\mathcal{G}_{\rm FE}$ isolates the native wait-only entangler. The other families test whether SNAP is useful as a final clean-up layer, as an interleaved correction after each entangling wait, or in both roles.

To attribute phase generation, we use two diagnostics already exposed by `cqed_sim`: `GateSequence.phase_decomposition(...)` for per-gate drift and SNAP phase tables, and `logical_block_phase_diagnostics(...)` to compare strict logical fidelity against a cavity-block gauge. Complexity is quantified by gate depth, parameter count, SNAP gate count, SNAP phase count, and total free-evolution wait time.

III. RESULTS

A. No-SNAP Free-Evolution Baseline

Figure 1 shows the fidelity-depth landscape across all screened families. The no-SNAP baseline improves steadily but slowly as additional free-evolution blocks are added. Its best refined case is `native_fe_b6`, which reaches fidelity 0.9639 with gate depth 20, total sequence duration 1858 ns, and total entangling wait time 1018 ns. The corresponding complexity counts are 48 continuous parameters and six entangling waits, with no explicit phase gates.

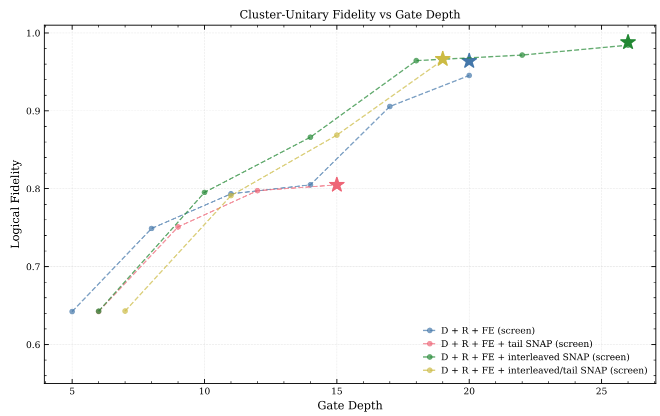

FIG. 1. Logical fidelity versus gate depth for the screened and refined families. The stars denote the best refined case in each family. Interleaved SNAP layers improve the achievable fidelity but do not dominate the pure native family at low depth.

The ablation data in Table III shows that free evolution is the essential entangling resource. Removing every FE block from the best native sequence drops the fidelity from 0.9639 to 0.1675, while the block-gauge fidelity falls to 0.2640. This is consistent with the primitive definitions: `Displacement` and `QubitRotation` provide mixing, but only `FreeEvolveCondPhase` produces qubit-conditioned phases in the restricted no-SNAP family.

B. SNAP-Extended Families

The extended families separate cleanly into one ineffective and two useful patterns. Tail-only SNAP layers are almost useless: the best refined `snap_tail_b4` case reaches only 0.8050, essentially matching the native four-block result. This means a single cavity-only phase layer added at the end cannot rescue a poor free-evolution decomposition.

Interleaved SNAP layers are different. The best refined `snap_interleaved_b6` case reaches 0.9881 with six FE blocks and six SNAP layers, compared with 0.9639 for the best native six-block sequence. The total entangling wait time is almost unchanged: 1027 ns versus 1018 ns. The price is complexity: gate depth rises from 20 to 26, the parameter count rises from 48 to 78, and the sequence now contains 24 explicit SNAP phase parameters. Interleaved SNAP therefore improves fidelity mainly by adding cavity-only correction freedom, not by replacing the entangler.

The hybrid family, which adds one extra tail SNAP on top of the interleaved structure, reaches 0.9665 with four FE blocks and five SNAPs. This is useful as a moderate-depth compromise, but it does not beat the best interleaved sequence in fidelity. The additional tail SNAP is therefore not the decisive ingredient.

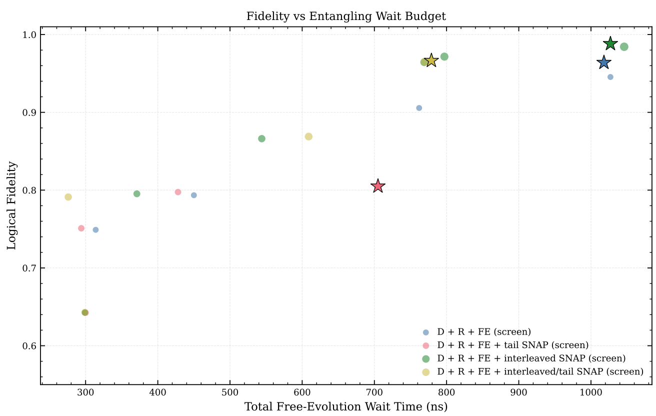

FIG. 2. Logical fidelity versus total free-evolution wait time. The high-fidelity region is entered by SNAP-assisted families at smaller wait budgets than the no-SNAP baseline, but the absolute best-fidelity point still uses roughly the same total wait time as the best native sequence.

C. Does SNAP Reduce Depth, Wait Time, or Only Add Redundancy?

The answer depends on which comparison point is used.

At the absolute best-fidelity endpoint, SNAP does *not* reduce the total wait time or depth. The best native and best interleaved cases both use six FE blocks and about 1.02 µs of total waiting, while the SNAP-assisted sequence is deeper and more parameter-rich. In that sense SNAP is not a shortcut entangler.

At practical fidelity thresholds, however, SNAP is clearly helpful. The first native case above 0.95 fidelity is `native_fe_b6`, with depth 20 and 1018 ns of waiting. By contrast, the interleaved SNAP family already reaches 0.9644 at `snap_interleaved_b4`, with depth 18 and only 769 ns of waiting. Relative to the no-SNAP threshold case, this is a 24.5% reduction in total entangling wait time and a reduction of two gates in depth. The same family reaches 0.9716 with five FE blocks and 797 ns wait, while the no-SNAP family never crosses 0.97 in the screened-plus-refined set.

Phase-budget and ablation data clarify the mechanism. The best native and best SNAP-assisted sequences accumulate nearly identical logical FE phase budgets: 5.78π and 5.83π, respectively. The best SNAP-assisted sequence adds an extra 1.14π of explicit logical cavity phase through its SNAP layers. When all SNAPs are removed from that sequence, the fidelity drops from 0.9881 to 0.4850 even though the FE budget is unchanged. When all FE blocks are removed, the fidelity falls further to 0.3223. Thus FE supplies the entangling phase, while SNAP supplies additional cavity-only phase correction that is independent and useful but not sufficient by itself.

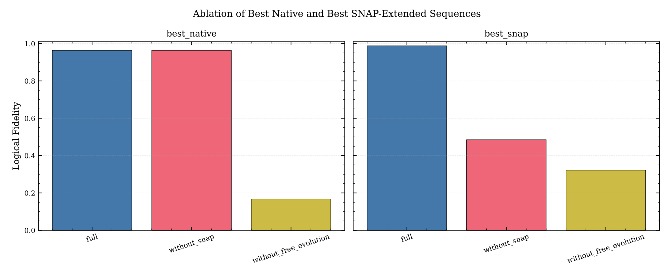

FIG. 3. Ablation of the best native and best SNAP-assisted sequences. Removing free evolution destroys both constructions. Removing SNAP from the best SNAP-assisted sequence also destroys high fidelity, confirming that FE and SNAP play distinct roles rather than one replacing the other.

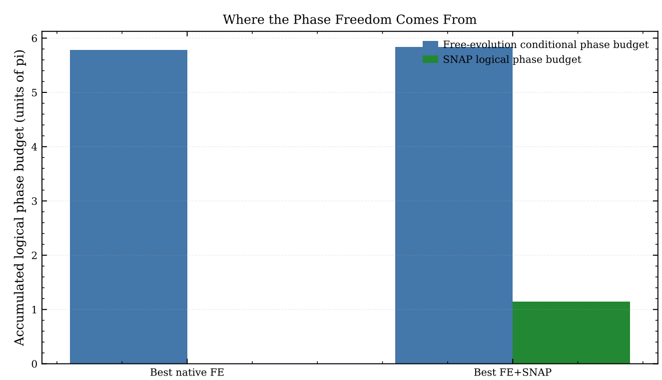

FIG. 4. Phase-budget attribution for the best native and best SNAP-assisted sequences. Both use roughly the same amount of qubit-conditioned phase from free evolution; the SNAP-assisted solution adds an extra cavity-only phase budget of about 1.14π.

TABLE II. Best refined result from each family.

Family	F	Depth	Wait (ns)	SNAPs	Params
Native FE	0.9639	20	1018	0	48
Tail SNAP	0.8050	15	705	1	39
Interleaved SNAP	**0.9881**	26	1027	6	78
Hybrid SNAP	0.9665	19	779	5	59

TABLE III. Ablation fidelities for the best native and best SNAP-assisted sequences.

Case	Full	Without SNAP	Without FE
Best native FE	0.9639	0.9639	0.1675
Best FE+SNAP	0.9881	0.4850	0.3223

IV. VALIDATION

A. Sanity Checks

The ablation study is the main sanity check. In the native family, eliminating FE destroys the decomposition, which matches the expectation that the no-SNAP family has no other entangling primitive. In the SNAP-assisted family, removing SNAP also destroys the decomposition, showing that the added explicit phase freedom is doing nontrivial work and is not merely idle redundancy. The near equality of strict and cavity-block-gauge fidelities for the best cases further shows that the remaining errors are not dominated by a removable block phase.

B. Convergence Analysis

The original winning sequences were first replayed without re-optimization at larger cavity truncation. The best native sequence dropped from 0.9639 at $N_{\mathrm{cav}} = 4$ to 0.9177 and 0.9188 at $N_{\mathrm{cav}} = 6$ and 8. The best SNAP-assisted sequence dropped from 0.9881 to 0.9473 and 0.9433 over the same truncation sweep. A follow-up improvement pass then re-optimized the two depth-6 winners directly at $N_{\mathrm{cav}} = 6$. This lifted the native family to 0.9234 and the interleaved-SNAP family to 0.9908 at the larger truncation. When those re-optimized winners are replayed coherently with the model-backed pulse-unitary backend at $N_{\mathrm{cav}} = 8$, the fidelities are 0.9238 (native) and 0.9867 (interleaved SNAP). The ranking is therefore much more stable than the original replay-only validation suggested, although the larger-truncation search has still not been repeated for every family and block count.

C. Noise Surrogate

The present `cqed_sim` build does not provide a direct noisy compiled replay for sequences containing `FreeEvolveCondPhase` and `SNAP`: the waveform bridge rejects both gates, and the gate-sequence `pulse` backend ignores `NoiseSpec` during state propagation. We therefore added a documented Lindblad surrogate that uses each gate's full pulse-unitary matrix and duration to construct a finite-duration effective Hamiltonian, then evolves the probe states with nominal $T_1 = 30\,\mu\mathrm{s}$, $T_\phi = 20\,\mu\mathrm{s}$, and cavity $T_1 = 200\,\mu\mathrm{s}$. At $N_{\mathrm{cav}} = 6$, the mean probe-state fidelity falls from 0.9190 to 0.8904 for the re-optimized native family and from 0.9910 to 0.9387 for the re-optimized interleaved-SNAP family. At $N_{\mathrm{cav}} = 8$, the same surrogate gives 0.8907 and 0.9351, respectively, so the qualitative ordering survives under the nominal coherence budget.

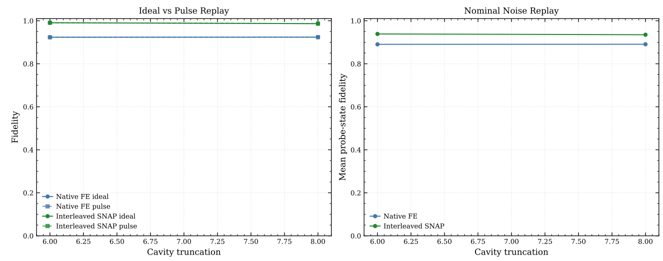

FIG. 5. Improvement-pass validation. Left: ideal versus coherent pulse replay for the re-optimized depth-6 native and interleaved-SNAP winners at $N_{\mathrm{cav}} = 6$ and 8. Right: nominal-noise Lindblad surrogate mean probe-state fidelity for the same two winners. The interleaved-SNAP sequence remains better at every tested truncation and under the nominal noise model.

D. Reference Comparison

There is no direct literature benchmark for this exact restricted gate-set comparison. The observed hierarchy is nevertheless consistent with the general bosonic-control literature: native waiting supplies the dispersive conditional phase, while SNAP provides cavity-only phase programmability [3, 4]. Our best FE+SNAP sequence remains below the stronger ideal-gate reference decompositions obtained previously in this repository with broader selective gate libraries, which is expected because the present study intentionally restricts the primitive set to isolate the role of free evolution.

V. DISCUSSION

The requested comparison yields a fairly nuanced answer. SNAP is not redundant, because the best interleaved FE+SNAP sequence loses half of its fidelity if the SNAP layers are removed while keeping the FE waits fixed. It is also not an alternate entangler, because removing FE is even more destructive and because the FE phase budget remains essentially unchanged between the best native and best SNAP-assisted solutions. The cleanest interpretation is that native free evolution carries the entangling burden and SNAP acts as an explicit local phase compiler around that entangler.

This interpretation also explains why the best-fidelity endpoint does not use less wait time with SNAP. To maximize fidelity, the optimizer still wants about six FE blocks and about $1\,\mu\mathrm{s}$ of total waiting. What SNAP buys is not a smaller FE phase budget, but the freedom to correct residual cavity phases and leakage-inducing mismatches produced by the displacement-rotation-wait structure. That is why the interleaved family crosses useful fidelity thresholds earlier in depth, even though its absolute best solution is more complex than the best native one.

The tail-only result is also informative. If SNAP were only needed as a final phase clean-up, then one tail SNAP should have helped substantially. It did not. The useful role of SNAP is therefore local and distributed: phase corrections must be applied while the decomposition is being built, not only after it is finished. By contrast, the extra tail SNAP in the hybrid family provides only a small improvement over the four-block interleaved construction, which suggests that much of the tail phase freedom is redundant once the interleaved layers are already present.

The larger-truncation extension materially strengthens this interpretation. Re-optimizing only the two depth-6 winners at $N_{\text{cav}} = 6$ already changes the quantitative story: the native family improves only modestly, from 0.9177 to 0.9234, whereas the interleaved-SNAP family jumps from 0.9473 to 0.9908. In other words, the larger truncation does not simply punish the SNAP-assisted sequence; it also reveals that the original $N_{\text{cav}} = 4$ parameter set left significant high-truncation performance on the table for the interleaved construction.

The noise-surrogate extension points the same way. Under a shared nominal coherence budget, the re-optimized interleaved-SNAP sequence still outperforms the re-optimized native one by about 4.5 percentage points in mean probe-state fidelity at both $N_{\text{cav}} = 6$ and 8. That does not settle the final hardware story, because the noisy result is currently a documented surrogate rather than a compiled cqed_sim replay, but it does remove the most obvious counter-argument that the deeper SNAP sequence must automatically lose once decoherence is included.

VI. CONCLUSION

The study goals were met. We isolated the native free-evolution family, compared it against several SNAP-extended gate sets, identified which contributions come from FE versus explicit phase gates, and quantified the trade offs in fidelity, depth, wait time, and complexity.

The main conclusions are:

- Native free evolution is essential. It is the only entangling resource in the restricted family, and removing it destroys the decomposition.
- SNAP is useful but not as a substitute entangler. Its benefit is to add cavity-only phase correction on top of essentially the same FE phase budget.
- At the best-fidelity endpoint, SNAP improves fidelity from 0.9639 to 0.9881 but does not reduce the required total wait time; it increases depth and parameter count instead.
- At practical thresholds, SNAP reduces resource requirements. The interleaved family exceeds 0.95 fidelity with four FE blocks and 769 ns of total wait, while the native family first exceeds 0.95 only with six FE blocks and 1018 ns of wait.
- Tail-only SNAP is largely redundant, whereas interleaved SNAP is the meaningful extension.
- A larger-truncation improvement pass strengthens rather than weakens the SNAP conclusion: re-optimizing the two depth-6 winners at $N_{\text{cav}} = 6$ yields 0.9234 for native FE and 0.9908 for interleaved SNAP, and the interleaved sequence remains ahead under both coherent replay and the nominal Lindblad noise surrogate.

VII. LIMITATIONS AND FUTURE WORK

A. Known Limitations

The first limitation is larger-truncation search coverage. The improvement pass re-optimized only the two depth-6 winners at $N_{\text{cav}} = 6$; it did not repeat the full family and block-count sweep at $N_{\text{cav}} \geq 6$. The updated numbers therefore establish that the native-vs-SNAP ranking survives a stronger truncation and even sharpens in favor of interleaved SNAP, but they do not yet prove that the original threshold frontiers remain globally optimal at larger cutoff.

The second limitation is replay realism. Coherent replay now exists via the model-backed pulse-unitary path, but the present `cqed_sim` waveform bridge still does not compile `FreeEvolveCondPhase` or `SNAP` into hardware-distorted pulses. The current pulse-level result is therefore a control surrogate, not a full compiled waveform validation.

The third limitation is noisy validation methodology. The direct gate-sequence `pulse` backend ignores `NoiseSpec` during state propagation for these FE/SNAP sequences, so the present decoherence analysis uses a documented local Lindblad surrogate derived from the per-gate pulse-unitary matrices and durations. This is more informative than a pure coherence-budget estimate, but it remains a surrogate rather than a native compiled noisy replay.

The fourth limitation is optimization scope. We searched structured ansatz families with modest multi-start budgets, not an exhaustive global optimum. In particular, a sparse-SNAP search that allows fewer than one SNAP per FE block at high depth may find a lower-complexity solution than the present best interleaved sequence.

B. Suggested Improvements

- **[P1 — HIGH]** Repeat the full family/block-count search at $N_{\mathrm{cav}} \geq 6$ (and eventually $N_{\mathrm{cav}} = 8$, $N_{\mathrm{tr}} = 3$), rather than re-optimizing only the two depth-6 winners. The current extension confirms the ranking for those winners but does not yet establish the larger- truncation threshold frontier globally.
- **[P2 — HIGH]** Replace the local Lindblad surrogate with a direct compiled noisy replay. This requires either waveform-bridge support for `FreeEvolveCondPhase` and `SNAP` or a native noisy Hamiltonian- slice runner for gate sequences.
- **[P2 — MEDIUM]** Search sparse interleaved SNAP patterns at larger truncation. The best solution still uses six SNAP layers, but the tail-only family is ineffective and the hybrid tail layer appears weak, so some SNAP placements are probably redundant.
- **[P2 — LOW]** Add a wait-budget constrained search instead of only a duration-weight sweep. In the current fixed-depth frontier, the optimizer compressed non-wait gate durations much more than the FE budget.
- **[P3 — LOW]** Compare the best FE+SNAP sequence to broader selective references and to GRAPE warm starts to see whether the present restricted decomposition is a useful compiler seed.

C. Open Questions

The main open question is whether the best FE+SNAP sequence genuinely needs one SNAP after every FE block, or whether a sparser pattern can retain most of the fidelity gain. A second question is whether the observed threshold improvement for SNAP persists after re-optimization at larger truncation. A third is whether the same qualitative conclusion holds for larger logical subspaces or other bosonic encodings, where the balance between native waiting and explicit phase programming may shift.

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[3] R. W. Heeres, B. Vlastakis, E. Holland, S. Krastanov, V. V. Albert, L. Frunzio, L. Jiang, and R. J. Schoelkopf, Physical Review Letters **115**, 137002 (2015).
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[5] A. Eickbusch, V. Sivak, A. Z. Ding, S. S. Elder, S. R. Jha, J. Venkatraman, B. Royer, S. M. Girvin, R. J. Schoelkopf, and M. H. Devoret, Nature Physics **18**, 1464 (2022).
[6] Shyam Shankar Quantum Circuits Group, cqed_sim: Circuit qed simulation framework (2025).

Appendix A: Detailed Results and Data

Table II gives the best refined result from each family. The full screened landscape is stored in `data/screen_results.json`, and the refined winners are stored in `data/refined_results.json`. The most useful threshold-level comparison is:

- Native FE first exceeds 0.95 fidelity at `native_fe_b6`: depth 20, wait 1018 ns, fidelity 0.9639.
- Interleaved FE+SNAP first exceeds 0.95 fidelity at `snap_interleaved_b4`: depth 18, wait 769 ns, fidelity 0.9644.
- Interleaved FE+SNAP first exceeds 0.97 fidelity at `snap_interleaved_b5`: depth 22, wait 797 ns, fidelity 0.9716.

These threshold points are saved in `data/threshold_summary.json`.

Appendix B: Reproducibility

1. Optimized Parameters

The full optimized gate parameters are provided in machine-readable form in `artifacts/best_native_sequence.json` and `artifacts/best_snap_sequence.json`. Each artifact lists every gate in order, including duration, optimized displacement amplitudes, qubit-rotation angles/phases, FE wait times, and every SNAP phase.

2. Waveform and Pulse Information

This study used ideal synthesis primitives rather than compiled pulses. The FE blocks therefore record optimized wait times, while SNAP records optimized phase vectors over the retained Fock levels. The nominal non-wait durations are 80 ns for displacement, 40 ns for qubit rotation, and 200 ns for SNAP.

3. Gate Sequence and Decomposition

The best native sequence contains six FE blocks and no phase gates. The best SNAP-assisted sequence contains six FE blocks and six interleaved SNAP layers. The phase-decomposition tables saved inside the JSON artifacts separate FE drift-phase contributions (`delta_phi`) from explicit SNAP relative phases (`relative_phases_rad`).

4. Modeling and Simulation Assumptions

Searches were run at $N_{\mathrm{cav}} = 4$ and $N_{\mathrm{tr}} = 2$ in the rotating frame defined by the shared cluster-state study helpers. Validation replayed the optimized sequences at $N_{\mathrm{cav}} = 6$ and 8 without re-optimization. The target was always the default `make_target("cluster", n_match=1)` convention with global phase ignored.

5. Reproduction Procedure

To reproduce the study:

1. Run `python studies/cluster_state_free_evolution_snap_comparison/scripts/run_study.py`.
2. Run `python studies/cluster_state_free_evolution_snap_comparison/scripts/run_improvement_pass.py` to reproduce the larger-truncation re-optimization and surrogate replay pass.
3. Inspect the saved JSON files in `data/` and the figures in `figures/`.
4. Verify the main claims by checking `study_summary.json`, `threshold_summary.json`, `ablation_results.json`, `validation_summary.json`, and `improvement_pass_summary.json`.

6. Saved Artifacts

The key artifacts are:

- `artifacts/best_native_sequence.json`: full optimized no-SNAP sequence.
- `artifacts/best_snap_sequence.json`: full optimized SNAP-assisted sequence.
- `artifacts/native_fe_b6_n6_reoptimized.json`: larger- truncation re-optimized native winner.
- `artifacts/snap_interleaved_b6_n6_reoptimized.json`: larger-truncation re-optimized interleaved-SNAP winner.
- `data/screen_results.json`: all coarse screened cases.
- `data/refined_results.json`: best refined case per family.
- `data/duration_frontiers.json`: fixed-depth duration-weight trade-off scans.
- `data/threshold_summary.json`: minimum-depth and minimum-wait solutions at useful fidelity thresholds.
- `data/ablation_results.json`: FE and SNAP ablation diagnostics.
- `data/validation_summary.json`: truncation replay metrics.
- `data/improvement_pass_summary.json`: larger-truncation re-optimization, coherent replay, and Lindblad-surrogate noise metrics.